Formatted Word report with embedded figures and source-data links

Overall Progress

本报告整理 TopoLink-CCI benchmarking 的完整收口状态。主线从 PDC Synthetic Truth 和 false-positive controls 出发，说明方法在受控场景下的 ranking behavior，再引入 Breast WTA 主 benchmark、A100 supplement、bounded appendix 和 Cervical WTA cross-dataset generalization。

当前 PDC publication queue 为 **0**，A100 无 active rescue process。报告口径已升级为 expanded 18-method benchmark：9 个 full success、8 个 bounded success、1 个 reproducible failure card、0 个 deferred candidate methods、0 个 pending/running。这里的 all_methods_accounted=true 表示所有提到的方法都有明确状态，不表示所有方法都成功。

FastCCC、SCILD、Copulacci 和 NicheNet 现在正式计入 expanded denominator，并且均已在 PDC 与 A100 完成终端化处理。FastCCC、SCILD 和 Copulacci 经 A100 retry 升级为 bounded subset result；NicheNet 保留 reproducible failure card，原因是 R dependency/API audit blocker。

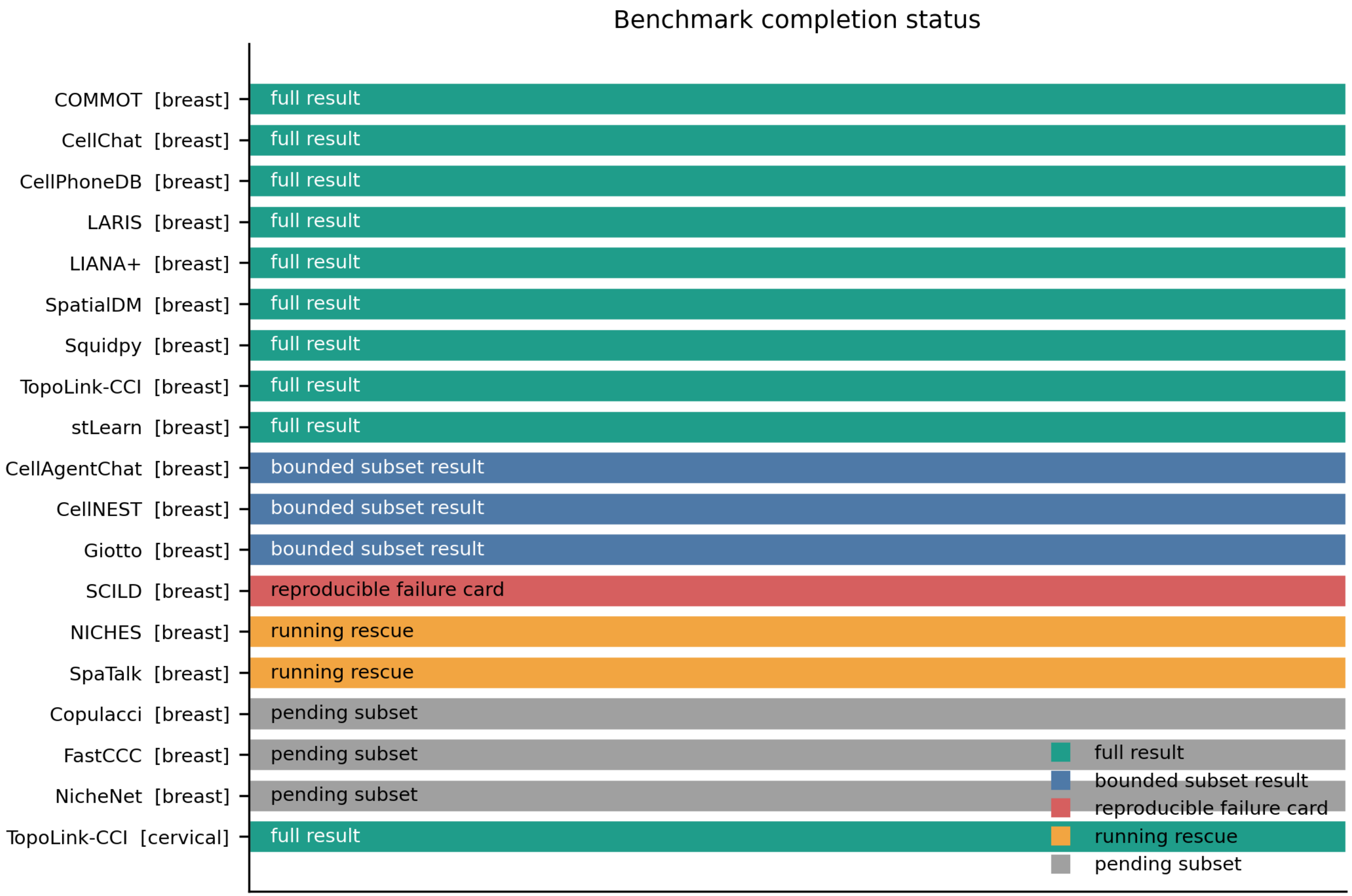
Key path: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\publication_benchmark_24h_20260511**
Key path: **/cfs/klemming/scratch/h/hutaobo/pyxenium_cci_benchmark_2026-04**
Key path: **/data/taobo.hu/pyxenium_lr_benchmark_2026-04**

Summary of Evaluated Methods

Dataset	Method	Status	Evidence tier	Phase	Rows	Hardware/source	Interpretation
Breast WTA	TopoLink-CCI	Full	Main	full common-db	1,319,600	PDC	Primary clean full-common result.
Breast WTA	CellPhoneDB	Full	Main	full common-db	reported in artifact	PDC	Reproducible non-spatial expression baseline.
Breast WTA	LARIS	Full	Main	full common-db	reported in artifact	PDC	Diffusion/spatial smoothing comparator.
Breast WTA	LIANA+	Full	Main	full common-db	reported in artifact	PDC	Spatial bivariate/multi-method comparator.
Breast WTA	SpatialDM	Full	Main	full common-db	reported in artifact	PDC	Spatial co-expression/null comparator.
Breast WTA	stLearn	Full	Main	full common-db	reported in artifact	PDC/A100	Spatial neighborhood permutation comparator.
Breast WTA	Squidpy	Full	Supplement	full common-db	1,319,600	A100	Permutation-style ligrec supplement.
Breast WTA	CellChat	Full	Supplement	LR-only full	31,592	A100	Pathway step disabled because of runtime bug.
Breast WTA	COMMOT	Full	Appendix	chunked full	724,604	A100/PDC	Transport/diffusion comparator; 16/16 chunks merged.
Breast WTA	Giotto	Bounded	Appendix	pilot50k	1,151,351	A100	Full limited by R Matrix/TsparseMatrix 2^31-1 sparse index boundary.
Breast WTA	SpaTalk	Bounded	Appendix	smoke20k	37,532	A100	66/66 smoke chunks completed; pilot50k had no usable output.
Breast WTA	NICHES	Bounded	Appendix	pilot50k	1,181,042	A100	Full stopped by memory safety gate.
Breast WTA	CellNEST	Bounded	Appendix	pilot50k	157,553	A100	Bounded GPU/source workflow result.
Breast WTA	CellAgentChat	Bounded	Appendix	pilot50k	1,143,543	A100	16/16 pilot chunks succeeded; full resource exceeded.
Breast WTA	SCILD	Bounded	Bounded appendix	smoke3k common-db	2,880 rows	A100	Official source-backed ligand-diffusion bounded retry succeeded: 3000 cells x 20 CCI pairs, peak RSS about 18.9GB.

Breast WTA	FastCCC	Bounded	Appendix	smoke20k	1,319,600	A100	A100 retry succeeded after PDC standardization failure; analytic non-spatial expression baseline.
Breast WTA	Copulacci	Bounded	Appendix	smoke20k	13,981	A100	Official source-backed copulacci.model2.run_scc bounded retry; 50k expansion stopped for dense adjacency memory risk.
Breast WTA	NicheNet	Failure card	Terminal card	env/API audit	reported in artifact	PDC/A100	R dependency/API audit blocker; downstream support method, not direct spatial CCI ranker.
Cervical WTA	TopoLink-CCI	Full	Cross-dataset	full common-db	2,404,971	PDC	Top hit: DSC2-DSG3 in differentiating tumor cells.

Figure panel 1. Benchmark completion status



Overview of full, bounded, and reproducible failure-card method states.

Figure: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\preview_completed_20260511\figures\fig1_completion_status.png**

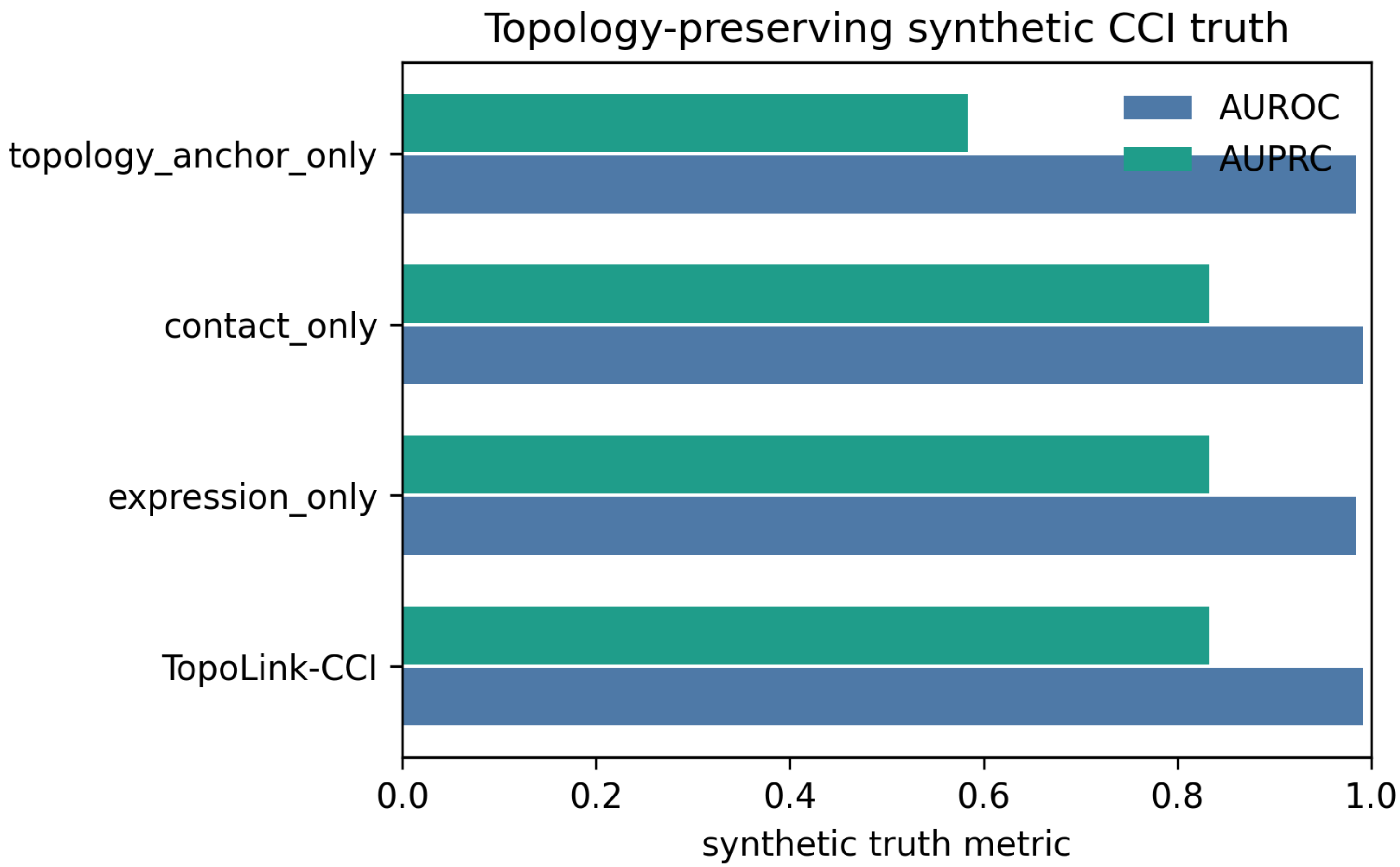
Source data: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\preview_completed_20260511\source_data\fig1_completion_status.tsv**

Validation & Controls

PDC Synthetic Truth 结果提供了真实 WTA benchmark 的前置可信度论证。完整 TopoLink-CCI 模型获得 **AUROC 0.9919** 和 **AUPRC 0.8333**。expression-only 为 AUROC 0.9839 / AUPRC 0.8333，contact-only 为 AUROC 0.9919 / AUPRC 0.8333，而 topology-anchor-only 虽然 AUROC 为 0.9839，但 AUPRC 降至 0.5833。

该结果说明 Topology Anchor 单独使用不足以稳定排序 positive CCI axes；完整模型通过 expression specificity 和 local contact 补充拓扑信息，提升 precision-recall ranking 的稳定性。

Figure panel 2. Synthetic Truth validation



TopoLink-CCI achieves AUROC 0.9919 and AUPRC 0.8333; topology-anchor-only loses precision-recall stability.

Figure:
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Source data:
D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\publication_benchmark_24h_20260511\pdc_collected\publication_benchmark_24h_20260511\synthetic_truth\tables\synthetic_truth_metrics.tsv

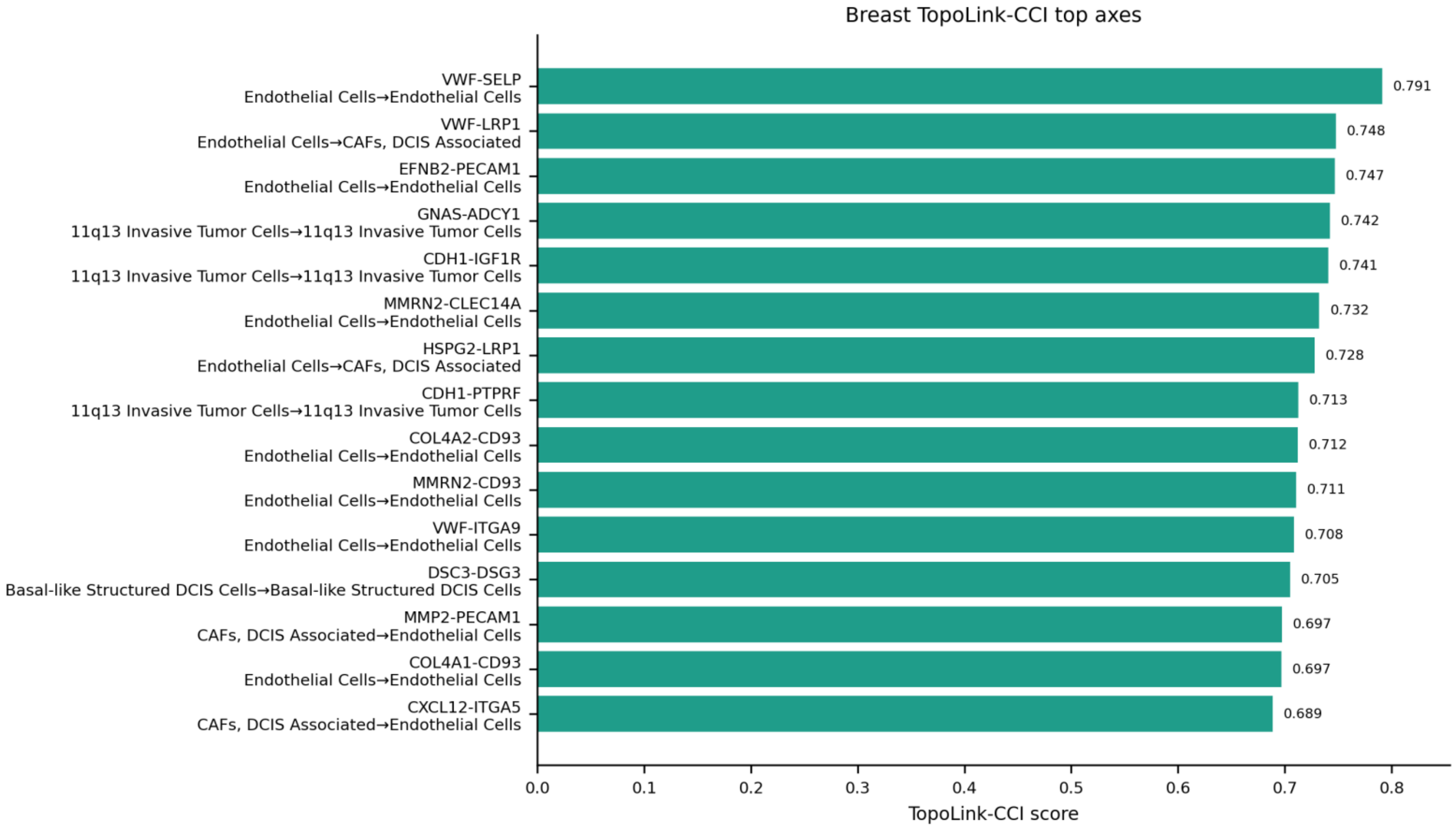
Core Benchmarking Results

Breast WTA common-db main comparison

Breast WTA 是主实验数据集。TopoLink-CCI Breast full result 产生 **1,319,600 rows**，构成 biological interpretation、classic CCI axis selection、validation 和 cross-method consistency 的主要输入。

TopoLink-CCI Breast artifact: **pdc_collected/pdc_20260426_1327/runs/full_common/pyxenium**

Figure panel 3. Breast WTA TopoLink-CCI top axes

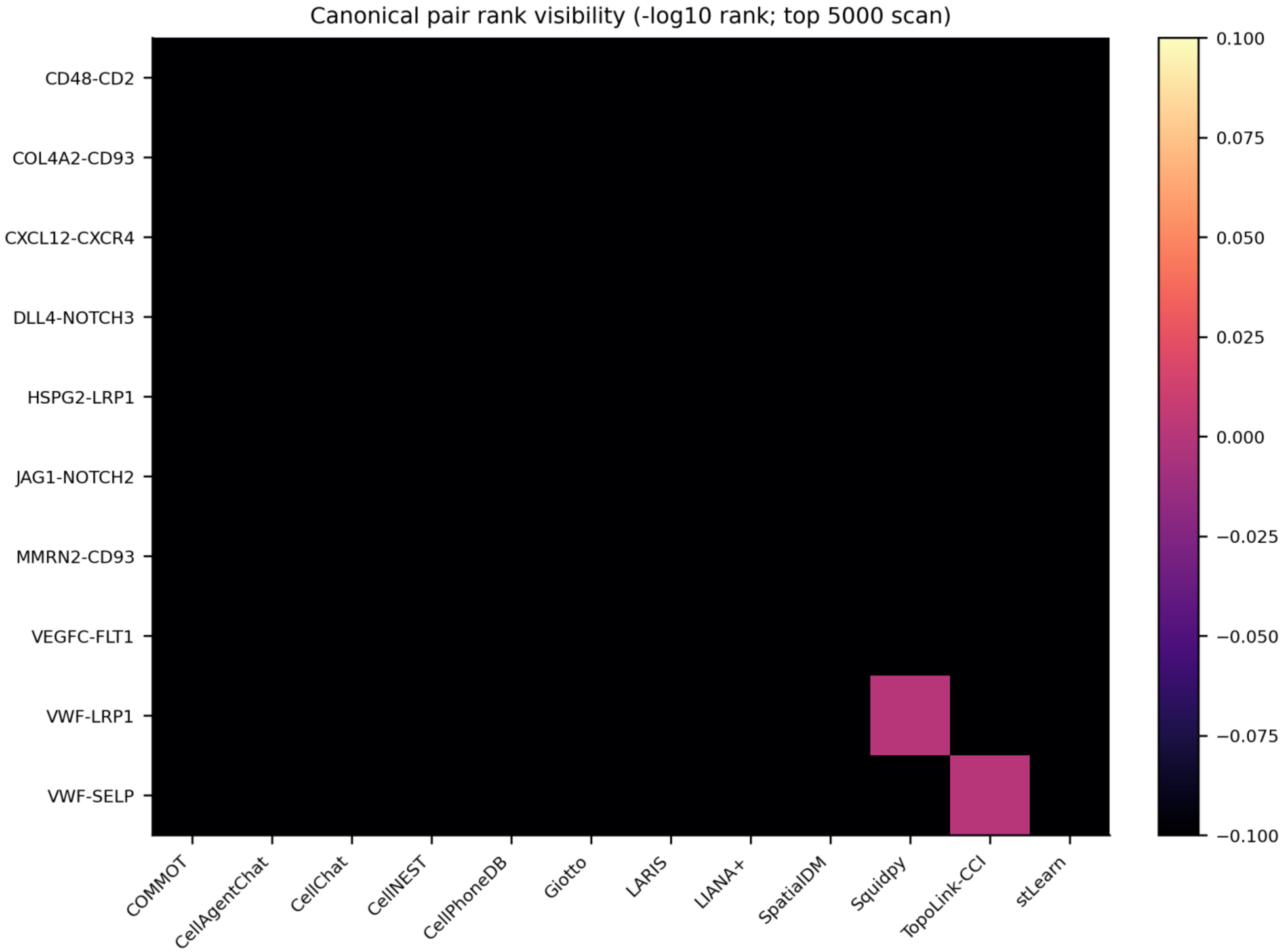


High-ranking Breast WTA axes emphasize vascular, stromal, immune and Notch-associated CCI themes.

Figure: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\preview_completed_20260511\figures\fig3_breast_topolink_top_axes.png**

Source data: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\preview_completed_20260511\source_data\fig3_breast_topolink_top_axes.tsv**

Figure panel 4. Canonical pair rank matrix



Canonical axis recovery is evaluated by within-method ranks rather than raw score comparison.

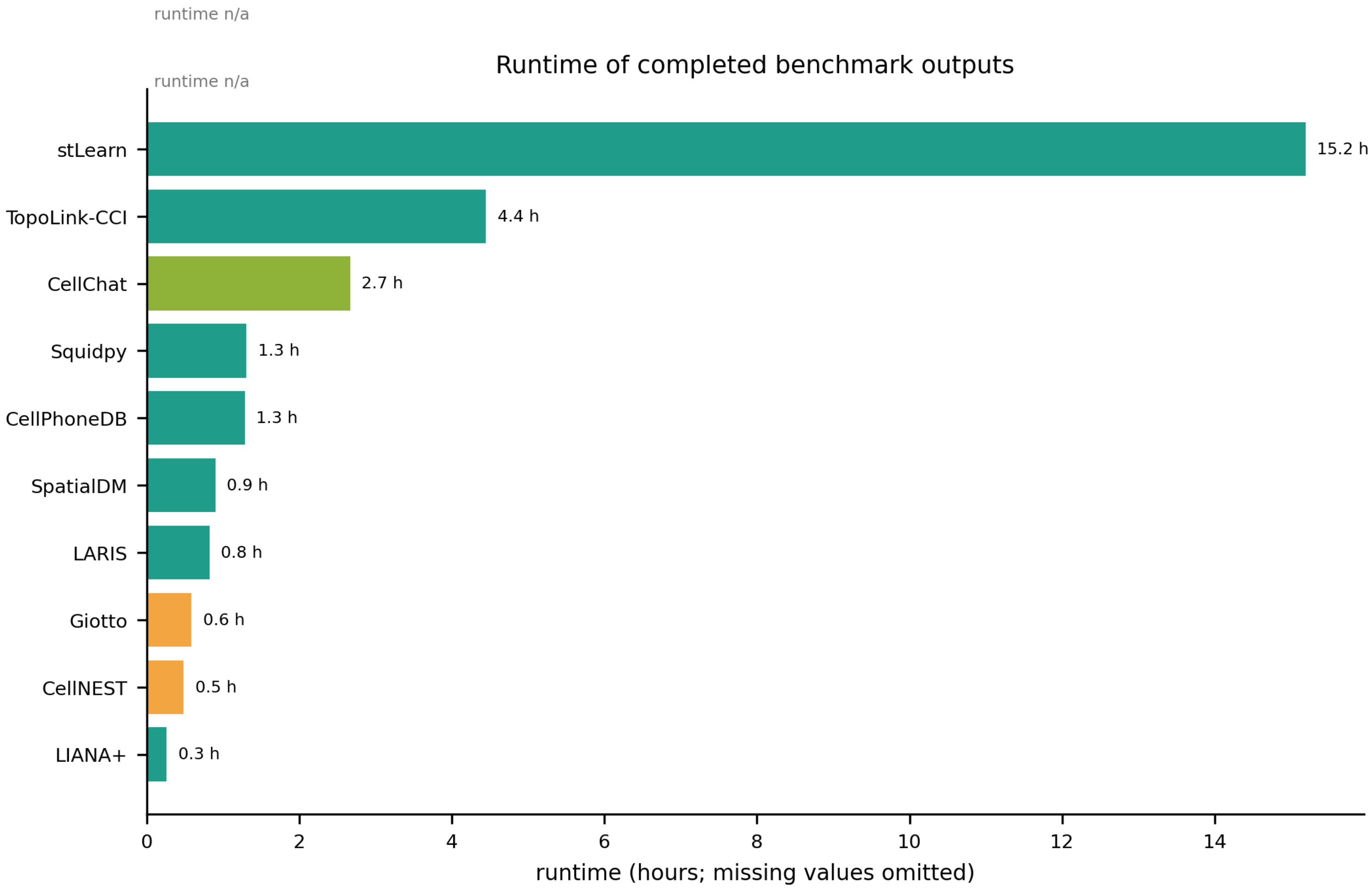
Figure: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\cross_method_comparison_20260511\figures\fig6_canonical_pair_rank_heatmap.png**

Source data: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\cross_method_comparison_20260511\source_data\fig6_canonical_pair_rank_heatmap.tsv**

A100 supplementary full and appendix runs

A100 supplement 提供 Squidpy full、CellChat LR-only full 和 COMMOT chunked result。bounded appendix 记录 Giotto、SpaTalk、NICHES、CellNEST 和 CellAgentChat 的 scalability/resource tradeoff。

Figure panel 5. Runtime and scalability

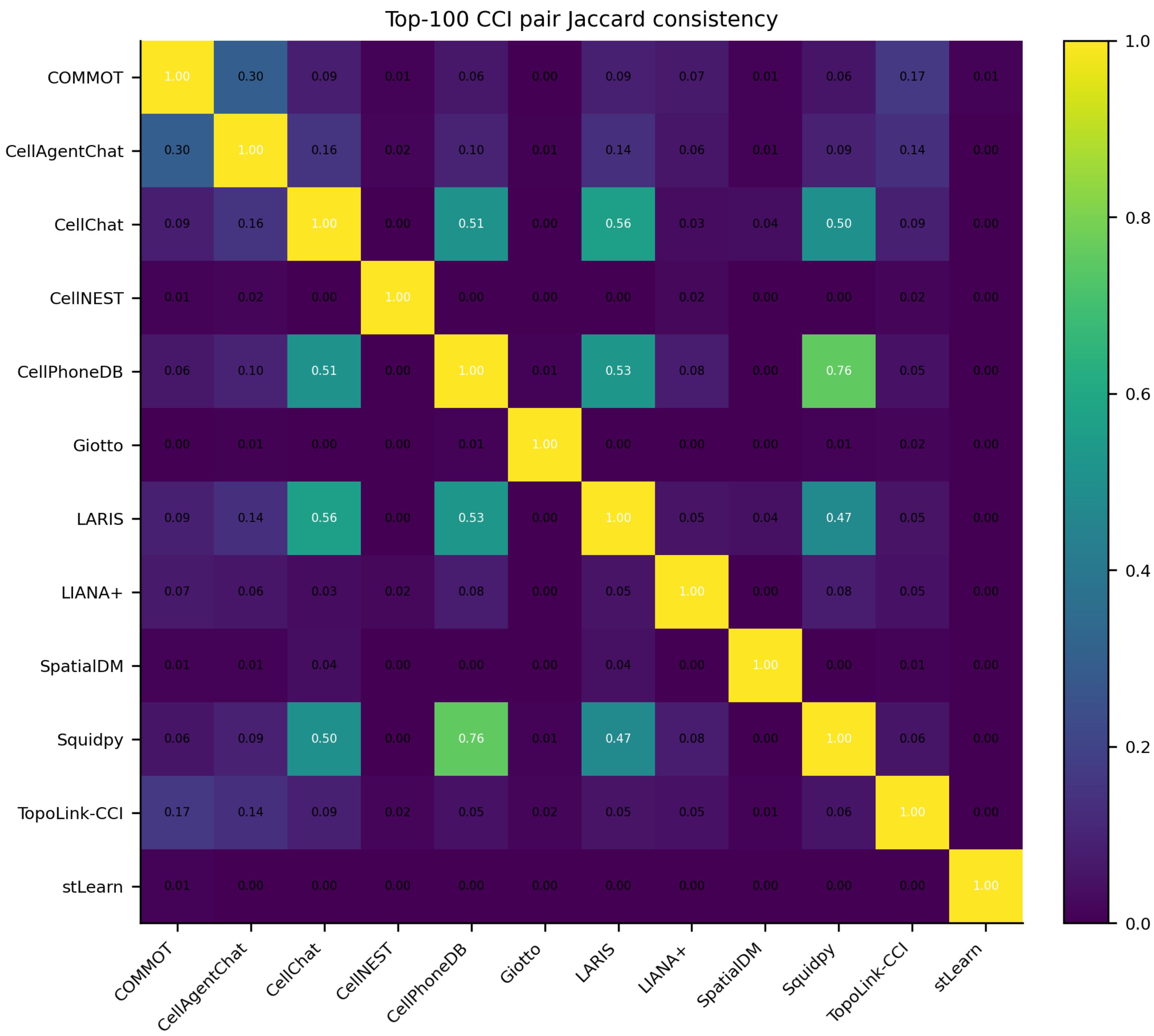


Engineering comparison across full and bounded runs.

Figure: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\cross_method_comparison_20260511\figures\fig1_runtime_hours.png**

Source data: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\cross_method_comparison_20260511\source_data\fig1_runtime_hours.tsv**

Figure panel 6. Top-100 pair Jaccard consistency



Cross-method consistency is summarized as overlap among top-ranked CCI axes.

Figure: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\cross_method_comparison_20260511\figures\fig3_top100_pair_jaccard_heatmap.png**

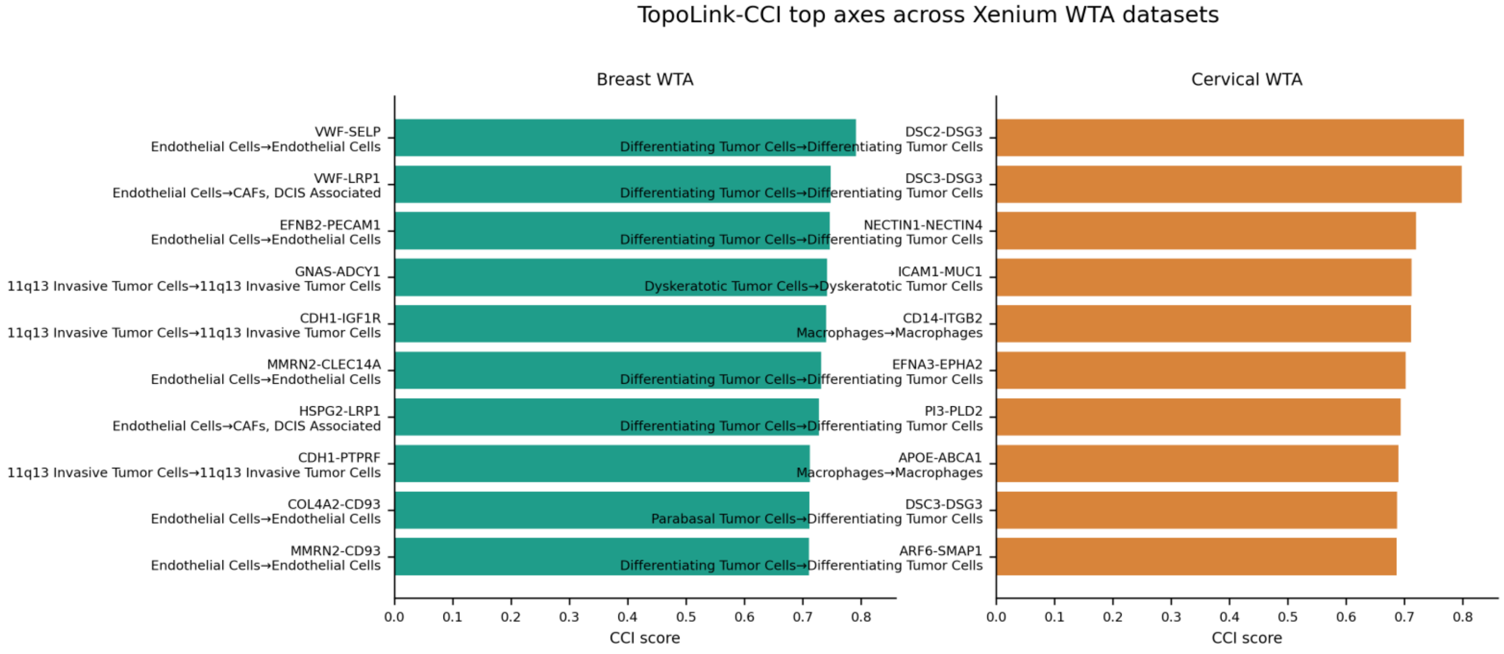
Source data: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\cross_method_comparison_20260511\source_data\fig3_top100_pair_jaccard_heatmap.tsv**

Cross-dataset generalization

Cervical WTA TopoLink-CCI full common-db run 生成 **2,404,971 rows**。其 top hit 为 **DSC2-DSG3 / Differentiating Tumor Cells -> Differentiating Tumor Cells**，与 Breast WTA 中 vascular/endothelial top axes 不同，支持 tissue-context-specific CCI prioritization。

Cervical artifact: **/cfs/klemming/scratch/h/hutaobo/pyxenium_cci_benchmark_2026-04/datasets/atera_cervical_wta/runs/full_common/pyxenium/pyxenium_standardized.tsv**

Figure panel 7. Breast versus Cervical generalization

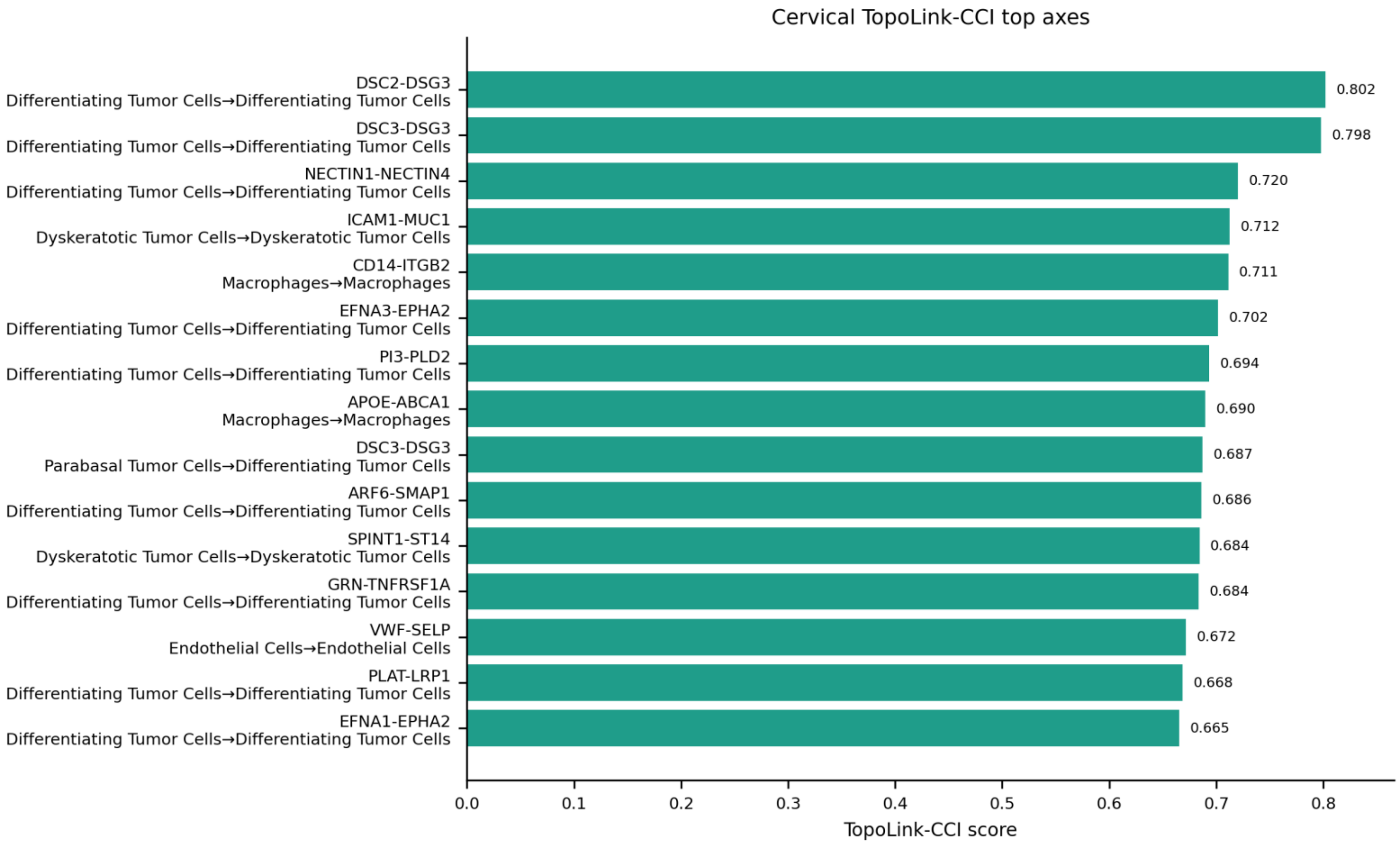


TopoLink-CCI recovers tissue-context-specific top axes across Breast and Cervical WTA datasets.

Figure: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\preview_completed_20260511\figures\fig4_breast_cervical_topolink_comparison.png**

Source data: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\preview_completed_20260511\source_data\fig4_breast_cervical_topolink_comparison.tsv**

Figure panel 8. Cervical WTA TopoLink-CCI top axes



Cervical WTA full common-db result contains 2,404,971 rows and is led by DSC2-DSG3.

Figure: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\preview_completed_20260511\figures\fig3b_cervical_topolink_top_axes.png**

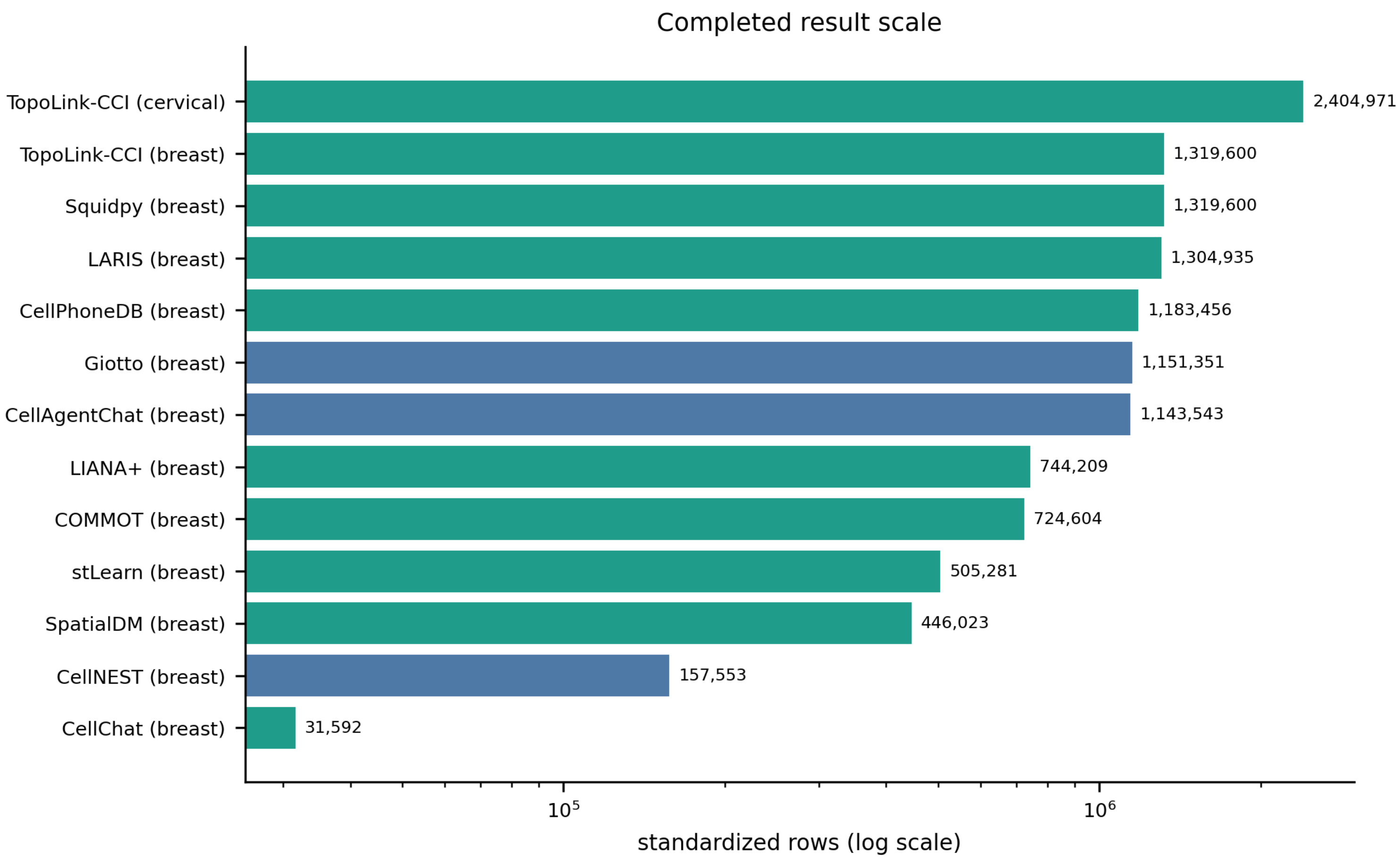
Source data: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\preview_completed_20260511\source_data\fig3b_cervical_topolink_top_axes.tsv**

Scope of Benchmark and Technical Limitations

本 benchmark 不把所有外部 CCI 方法强行推进到 whole-dataset full run，而是为每个方法保留 full result、bounded subset result 或 reproducible failure card。SpaTalk smoke20k 完成 **66/66 chunks**；NICHES pilot50k 完成 1,181,042 rows；Giotto pilot50k 完成 1,151,351 rows。这些结果说明真实方法链路可运行，同时也暴露了大规模 WTA 下的内存、sparse index 和 runtime scalability tradeoff。

SCILD 已在官方 source 被重新确认后通过 A100 专用 Python 3.11 环境升级为 bounded subset result：3000 cells × 20 CCI pairs，2,880 rows，峰值内存约 18.9GB。Copulacci 已通过官方 source-backed copulacci.model2.run_scc workflow 升级为 bounded subset result：20k cells × top 200 CCI pairs × top 80 celltype groups，13,981 rows；50k 扩容因 dense spatial adjacency memory risk 停止，因此不进入 full comparison。NicheNet 是 expanded 18-method benchmark 中仍保留的 reproducible failure card；它在 PDC 和 A100 均受 R dependency/API audit 约束，并且其定位更接近 downstream receiver-response support，而不是直接 spatial CCI ranker。FastCCC 已通过 A100 retry 产生 20k bounded standardized output。

Figure panel 9. Result row counts

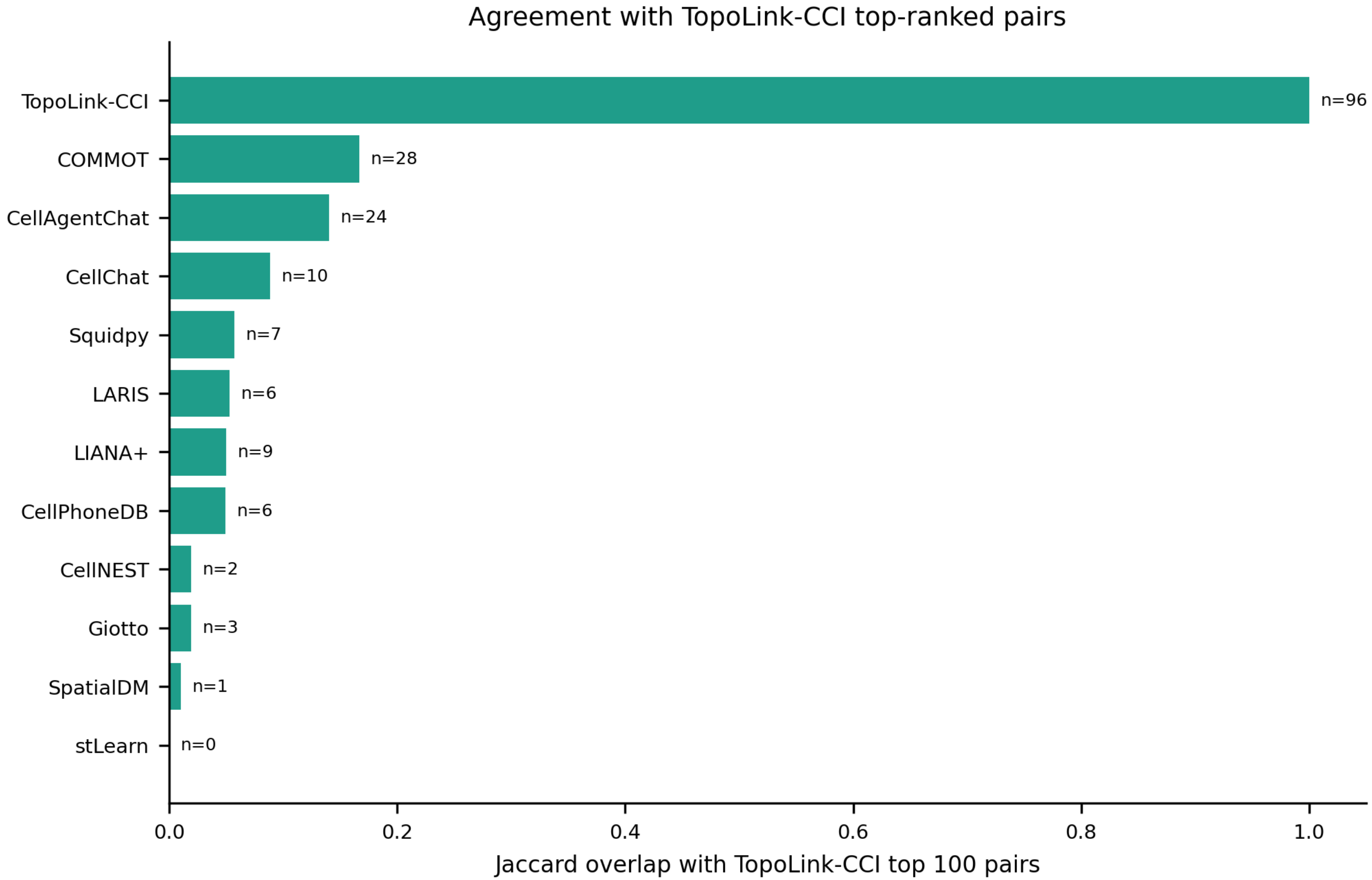


Full and bounded outputs differ in scale and should not be collapsed into a single evidence tier.

Figure: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\preview_completed_20260511\figures\fig2_result_row_counts.png**

Source data: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\preview_completed_20260511\source_data\fig2_result_row_counts.tsv**

Figure panel 10. Overlap with TopoLink-CCI



Overlap with TopoLink-CCI is interpreted as theme-level support, not absolute ground truth.

Figure: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\cross_method_comparison_20260511\figures\fig5_overlap_with_topolink.png**

Source data: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\cross_method_comparison_20260511\source_data\fig5_overlap_with_topolink.tsv**

Methodological Discussion

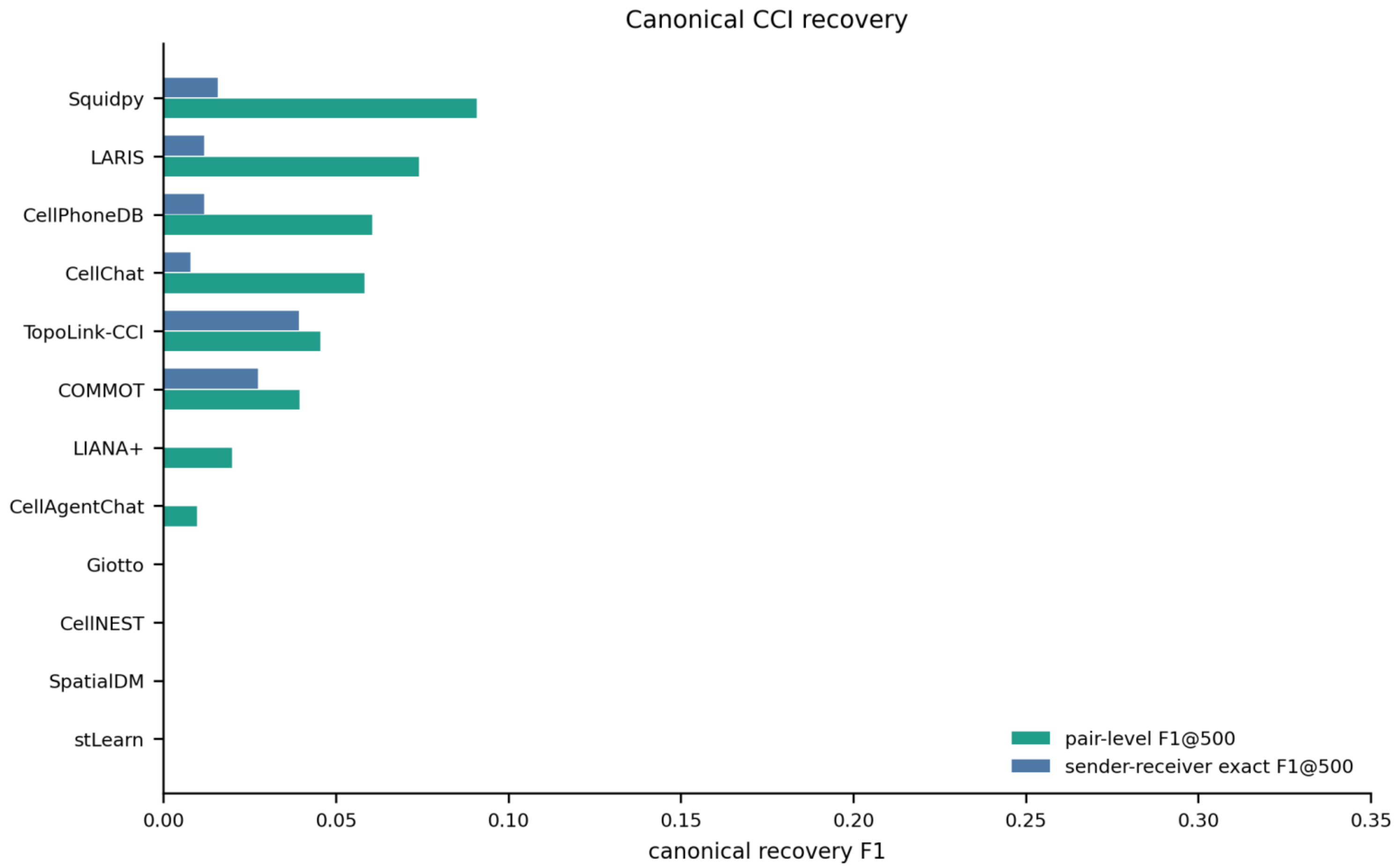
本 benchmark 不直接比较不同方法的 raw score，因为各方法的 score 定义不一致。评价体系采用 rank、canonical recovery、cross-method consistency、false-positive controls、Synthetic Truth 和 cross-dataset generalization。

F1 只用于 Synthetic Truth 或预定义 canonical truth set；ARI 不作为主指标，因为当前输出是 interaction-axis ranking，而不是统一 clustering assignment。

本研究建立的评估框架旨在为真实复杂组织切片中的 CCI 优先级排序提供可靠依据，而非单纯追求合成数据集上的指标拟合。

TopoLink-CCI 应被表述为 topology-guided spatial CCI hypothesis prioritization framework。所有结论均为 computational evidence，不应被解读为单靠 CCI_score 证明蛋白结合、因果通讯或功能效应。

Figure panel 11. Canonical recovery F1

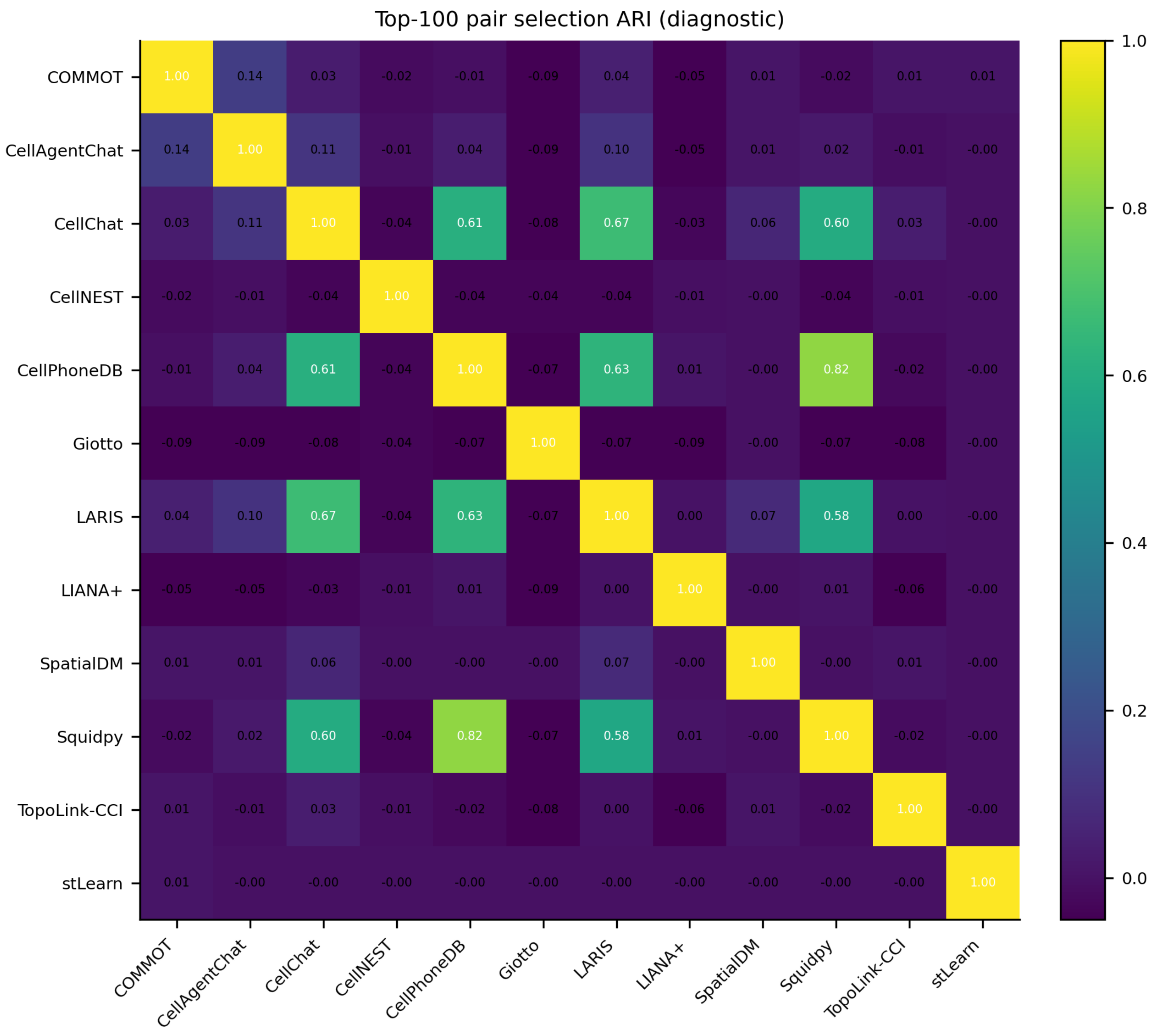


F1 is used only for predefined canonical truth sets, not global real-tissue CCI ranking.

Figure: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\cross_method_comparison_20260511\figures\fig2_canonical_recovery_f1.png**

Source data: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\cross_method_comparison_20260511\source_data\fig2_canonical_recovery_f1.tsv**

Figure panel 12. Exploratory ARI of top-k binary selections



ARI is shown as an exploratory top-k selection similarity metric, not a main endpoint.

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Source data: **D:\GitHub\pyXenium\benchmarking\cci_2026_atera\results\cross_method_comparison_20260511\source_data\fig4_top100_binary_selection_ari_heatmap.tsv**